

MPLPB Image

A Signed, Retrieval-Bounded Wall for Generated and Captured Pictures

Mitchell D. McPhetridge

Independent Researcher

Field	Value
-----	-----
Document ID	MPLPB-IMAGE-016
Category	Specification / Reference Implementation
Updated	2026-09-28T00:00Z v2
Status	current
Scope	The image cut of an MPLPB local web: how a studio and an importer write signed pages instead of anonymous files; how a picture's kind (generated, captured, unknown) is kept apart from its provenance; how evidence, perceptual matching, attestation, and a bounded learner decide what a reader may treat as a photograph
When to use	Keeping generated and found pictures where a later reader, person or model, must not treat a generated picture as a photograph; deciding what an image folder is allowed to claim

Related. Networked MPLPB (github.com/mitchell-d00/networked-mplpb): signed descriptions, origin depth, recorded ratification, keys outside corpora. MPLPB at Swarm and Enterprise Scale (MPLPB-SWARM-013 v3): epistemic laundering, deployment profiles, the ablation discipline. Smart Local MPLPB (MPLPB-SMART-011 v1): the retrieval-bounded reader. MPLPB and the Cost of Visibility (MPLPB-COST-012 v2). The Local Mirror (MPLPB-LOCAL-008 v4). Continuity Without Memory (2026).

[Back to. Main Index](#) > [Specification / Architecture Sub-Index](#)

Abstract

A folder of pictures remembers pixels and forgets how they were made. A later session sees a PNG and a timestamp. It cannot see that a model made it, from which prompt, or whether anybody ever said it was real. Monday generated. Thursday treats the file as a found object. That is the terminal-fact failure from Continuation Must Preserve Revisability, applied to pixels.

MPLPB Image keeps every picture as a page. Each page records two things that folders merge and this design keeps apart: what the pixels are (kind: generated, captured, or unknown) and where the record came from (origin, origin depth, and a signature). The studio writes generated pictures and signs them at ingest. An importer brings outside pictures in and assigns kind from evidence only: identical bytes or a perceptual match to a known generation, or the file declaring its own generation.

Trust is deliberately asymmetric. A claim that a picture is generated is accepted from evidence, because the error it risks is caution. A claim that a picture is captured is never accepted from the file. Camera metadata is a hint anyone can type. A picture becomes captured only when a named person attests it in their own words, signed and logged.

A small learner trains on attested pictures and scores unknown ones. Its score is advisory. It never changes a picture's kind, it never trains on its own output, and it reports by feature group so it cannot hide that it has only learned the file format.

Every record, every log event, the manifest fingerprint and the model are Ed25519-signed by a key held on the machine and never inside the seed. Twelve checks re-derive what can be re-derived: evidence from bytes, hashes from files, signatures from keys. A validity check needs only the seed. A trust check needs the machine's own choice of keys, and the external profile requires it.

This paper specifies the two axes, the evidence rules, perceptual matching, signatures and trust, the learner, the checks, the profiles, the write-backs, and the tests that have and have not been run. It does not claim to detect generated pictures. It claims to keep track of what is known about them, and to refuse to invent the rest.

1. The gap this fills

A generator toy (stdlib HTTP proxy, one wall, key in memory, bytes saved as received) answers the request question well and the reader question not at all. Three defects belong to whatever reads the folder later:

□ retrieval without provenance — a PNG with no document identity

□ model bypass — a later agent captions the file from its own weights

- local/public confusion — a generated file cited as if it were a photograph

A fourth appears as soon as the folder takes in pictures from outside:

- kind confusion — nobody can say which pictures were made and which were taken, and the absence of a generator tag is read as a camera

No amount of request quality in the studio prevents any of these. MPLPB Image is the reader discipline for the folder.

2. What a front end is allowed to do

The asymmetry is Smart Local §2. A wall that says I don't have that picture is mildly annoying. A wall that invents a caption indistinguishable from a stored record is corrosive, because after a few undetected instances the record on every other picture stops meaning anything.

ask() therefore has no path that describes a picture without a catalog hit. When nothing matches it lists the scopes that exist and stops. When something matches it returns the stored prompt or import title, labelled as such, with the picture's kind, origin, depth, status, and a citation line. It never returns a description of a scene. Matching is on whole words.

3. Two axes

Folders collapse two questions into one file. This design keeps them apart.

Kind answers: what are these pixels?

kind	how a picture gets it
generated	signed ingest by this studio; identical bytes to, or a perceptual match of, a known generation; or the file declares generation (IPTC, C2PA AI assertion, generator tag, diffusion parameters)

captured	a named person attests it; or it is a byte-identical copy of a picture someone attested
unknown	everything else

Origin answers: who stands behind this record, and how far is it from a person?

origin	depth	meaning
-----	-----	-----
machine	>= 1	generated; nobody has reviewed it
ratified	0	generated; a named person approved it as an illustration
human	0	captured; a named person attested it
unknown	>= 1	brought in from outside with no attestation

The combinations are closed. Generated is machine or ratified. Captured is human at depth 0. Unknown is never depth 0. Ratification never changes kind: an approved illustration is still generated and is never served as a photograph. Networked MPLPB's origins (human, machine, ratified) carry over unchanged; unknown is added because an importer, unlike an author, often does not know.

3.1 Asymmetric trust

Evidence of generation is accepted when it is strong. The cost of wrongly calling a photograph generated is that a real picture is served with a caution label. That is the cheap error.

Evidence of capture is never accepted from the file. EXIF make, model, exposure and GPS can be written by anyone with a text editor, and the absence of a generator tag means only that the tag is absent. A picture is captured only by attestation or by being byte-identical to an attested picture. The cost of wrongly calling a generated picture captured is the failure this whole design exists to prevent, so nothing a file says about itself can cause it.

The cheap error is cheap only if it can be corrected. When a picture is generated because of what the file declares and for no other reason, a named person may attest it as a capture over that declaration, but only by asking for it explicitly (§12). The overridden signals stay in the signed record, in the signed log and on the page. A match to a known generation is evidence about the pixels, not a claim the file makes, and is never overridable.

4. Layers

Layer	Modules	Owns
-----	-----	-----
1. Seed	corpus.py, validate.py	what a picture page is; whether the wall is well-formed
2. Evidence	evidence.py, pixels.py	what the bytes say about themselves; perceptual hash; content statistics
3. Kind	classify.py	generated / captured / unknown, from evidence only
4. Provenance	keys.py, ed25519.py	signatures, signer registry, local trust store
5. Learner	learn.py	advisory scores from attested labels
6. Retrieval	retrieve.py, cite.py	finding pages; citations that carry kind and signer
7. Decision	profiles.py	how a deployment may serve a picture
8. Write-back	lifecycle.py, studio	ingest, import, retire, ratify, attest

The seed never holds a secret key and never holds the API key. The studio loads the signing key from the machine at start and never serves it.

5. Ingest and import

Ingest (the studio, on a 200 from /v1/images/generations):

1. Decode b64_json, or fetch url over https only, capped at 64 MB.
2. Refuse anything whose bytes are not PNG, JPEG, or WebP by signature.
3. Read evidence and compute the perceptual hash, without changing a byte.
4. Allocate IMG-YYYYMMDD-XXXXXX, unique within the catalog.
5. Write the bytes unchanged under wall/, named by what they are.
6. Write the record: kind=generated, basis=signed-ingest, origin=machine, depth 1, prompt and prompt hash, model, evidence summary, dHash.
7. Sign the record, append a signed hash-chained lineage event, re-render the wall and log, rebuild and sign the manifest.

Import (the CLI, for a picture from anywhere):

1. The same byte checks, evidence and hash.
2. Refuse bytes already on the wall as a current page.
3. Decide kind by \$3 from the catalog and the evidence.
4. A generated import is machine at depth ≥ 1 , inheriting the depth of what it matched; copying never resets depth. An unknown import is origin unknown at depth 1. A captured import (a byte-identical copy of an attested picture) carries the attestation over.
5. If a learner exists, attach its advisory score to an unknown import.
6. Sign, log, render, manifest, as for ingest.

check does steps 1 and 3 and stores nothing.

6. Evidence

evidence.py reads container metadata from PNG (IHDR, tEXt, zTXt, iTXt, eXif, iCCP, caBX), JPEG (APP1 EXIF and XMP, APP2 ICC, APP11 JUMBF, quantization and frame headers) and WebP (VP8X, VP8L, VP8, EXIF, XMP, ICCP, C2PA). It never decodes pixels and never modifies bytes. Every read is bounded, and a truncated or malformed file produces parse errors, not exceptions; a fuzz test mutates files and requires the parser to survive.

Each signal is sorted by what it points to and how strong it is.

signal	points to	strength
-----	-----	-----
IPTC digital source type: trained or composite algorithmic media	generated	declared
C2PA manifest asserting AI generation	generated	declared
a known generator named in EXIF software, make or model, XMP creator tool, C2PA claim generator, or a PNG text key generators write (Software, Source, Generator, parameters, prompt, workflow)	generated	declared
diffusion parameters or a node workflow embedded in PNG text	generated	declared
IPTC digital source type: digital capture, film, print	captured	hint
camera make or model; exposure; GPS	captured	hint
C2PA manifest without an AI assertion	captured	hint
image-API standard size, no camera data	generated	hint

Declared generation sets kind. Hints set nothing; they are shown on the page and given to the learner.

Generator names are looked for only in fields whose job is to name the software that wrote the file. Captions, descriptions, titles, comments and author fields are free text written by people, and are never searched. v1 searched them, and a photograph captioned "Mi imagen del puerto", "the OpenAI office" or "luminous flux at dusk" was recorded as declaring its own

generation. Patterns for names that are also ordinary words (Imagen, FLUX) require a version or vendor form. A JUMBF box counts as C2PA only when its label says c2pa.

C2PA manifests are detected and not validated. This implementation does not check their signatures, and every C2PA field it reports says `verified: false`. A signed C2PA manifest from a trusted issuer is stronger evidence than anything here; validating one needs a certificate chain this design does not carry.

The evidence summary is stored in the signed record and re-derived from the bytes at every validation (I.11), so a record cannot claim a camera the file does not name, even when the key holder signs the claim.

7. Perceptual matching

Networked MPLPB states that verbatim copying does not reset depth. Without matching, that rule could only be contained: a copied PNG kept its record only while it kept its card. A picture resized for a message or re-encoded by a platform arrived as a stranger.

`pixels.py` computes a 64-bit difference hash over a 128-pixel thumbnail. Import compares it with every picture in the catalog, current and retired. A distance of 8 bits or less counts as the same picture. A resized, re-encoded copy of a studio generation is therefore imported as generated, with basis `perceptual-match`, naming the page it matched and inheriting its depth. For resizing and recompression, `FM-N7` is detected, not only contained.

Three limits are stated because each was found in testing:

□ Near-flat pictures collide. A clear sky and a generated gradient hash alike. A hash with fewer than 8 or more than 56 set bits is not used for matching; exact byte matches still count.

□ A perceptual match to an attested capture does not make an import captured. Only byte identity carries an attestation over. An edited copy of a photograph is unknown, with the resemblance reported as a conflict.

- Cropping, flipping, heavy editing, and a deliberate adversary defeat a `dHash`. A match means very probably the same picture. No match means nothing.

PNG is decoded with the standard library under two hard limits: a 40-megapixel cap read from the header, and a decompressed-size cap enforced while inflating, so a small file that inflates to gigabytes is refused. Pillow, if installed, decodes JPEG and WebP. Both paths share one exact area-average downsampler, so a PNG hashes identically with or without Pillow and a seed validates the same on either machine. An earlier draft used Pillow's own `resize` and disagreed with the standard-library path by six bits on a noisy picture; a test caught it.

8. Signatures and trust

Each machine holds an `Ed25519` key under `MPLPB_IMAGE_HOME`, created on first use with mode `0600`, outside every seed. The node ID is `N-` and the first sixteen hex digits of the `SHA-256` of the public key, as in Networked MPLPB, and the signature code is Networked MPLPB's pure-Python `RFC 8032` implementation carried unchanged.

What is signed: every record (over its identity, paths, hashes, kind, basis, origin, depth, evidence, and the names and times of any ratification or attestation); every lineage event, each naming the hash of the one before; the manifest fingerprint; and the learner model. The seed holds only public keys, in `_net/signers.json`.

Validity and trust are different questions and are answered separately.

question	asks	answered by
valid	does each signature verify against the key the seed names?	the seed alone
trusted	is that key one this machine chose to trust?	the local trust store only

A write that finds signers.json naming a different key for its own node ID refuses rather than repairing it. An earlier draft of this implementation silently rewrote the entry and erased the evidence; a test caught it.

8.1 Threat model

attacker	outcome
edits a record, page, log entry, or file without a key	signature or chain fails (I.9); page and catalog disagree (I.3); manifest (I.8)
relabels a generation as a capture without a key	signature fails; no attest event (I.5); external refuses
holds a key and signs a false camera into the evidence	evidence re-derived from bytes (I.11) disagrees
writes a secret key into a seed	I.10 fails
re-signs everything with their own key	valid; untrusted; validate --strict and the external profile refuse
swaps the public key for a node ID	the ID no longer hashes from the key (I.9); writes refuse
holds a trusted key and attests a generated picture that carries no generation evidence	succeeds. Attestation is a signature, not a proof; the attester's name is on the record and in the log
strips metadata and crops a generated picture	arrives as unknown. Nothing here can tell, and nothing here claims to

The last two rows are the limits. A trusted key that lies is outside what signatures can fix; the design makes the lie attributable and permanent in the log. A laundered picture with its evidence removed is outside what evidence can fix; the design keeps it unknown rather than guessing.

9. The learner

The request behind this section is to learn the difference between real and generated. A learner that writes its answer onto a card is a truth engine, and a laundering channel: a guess becomes a document, and the next reader retrieves it as a finding. Swarm MPLPB §5 is the same mechanism with text. So the learner is bounded by three rules.

Labels come only from attested pages: pictures the studio signed at ingest (generated) and pictures a named person attested (captured). Declared metadata, perceptual matches, and the learner's own earlier scores are never labels. A model trained on its own guesses grades itself.

Features are recomputed from the bytes at training time, after the bytes are checked against their signed hash. A picture whose bytes changed is skipped and named, not learned.

The output is advisory. It is a probability with the model hash beside it. It never changes a picture's kind, origin or depth, and an attested capture drops any score it had.

The model is a logistic regression in the standard library, over three feature groups:

group	features
format	container, API-standard size, squareness, pixel count
metadata	EXIF presence, camera, exposure, GPS, original date, software, ICC, XMP, PNG text, JPEG quality, progressive

	scan, chroma subsampling
content	compressed bytes per pixel, luminance spread, Laplacian mean and variation, Laplacian-to-gradient ratio, clipping, saturation, colour diversity

Declared generation signals are excluded. The rules in §3 already act on them, and including them lets a model score well by re-learning a rule.

Cross-validation is stratified and reported three times: all features, format and metadata only, content only. If format and metadata alone score within three points of all features, the report warns that the model is reading file type, size and EXIF, which stripping or re-encoding defeats. That warning is the point of the ablation. The first synthetic test set separated perfectly on format features because noisy pictures compress worse; compressibility was moved to the content group, where it belongs.

The model file is signed. I.12 checks that every training entry names an attested page whose bytes still match.

What the learner has shown: on synthetic pictures, it separates smooth renders from noisy textures on content alone and warns when a set can be separated by format alone. What it has not shown: anything about real generators against real cameras. §16 states what that needs.

10. Checks

check	asks
I.1	index.html exists and carries a document ID
I.2	the wall lists every current picture and no retired one
I.3	every catalog entry has its page, and the page agrees with the catalog on ID, status, kind, basis, origin, depth, and signature
I.4	document IDs are well-formed and unique
I.5	kind, basis, origin and depth form an allowed combination (§3); ratification and attestation have matching lineage records; a capture whose file declares generation carries an override that the log records identically; matches name real pages of the right kind; no picture page sits outside the catalog
I.6	the bytes each page names exist and match their signed hash; no image anywhere in the tree lacks a page
I.7	every link resolves to a file inside the root
I.8	the manifest lists exactly the tree with current hashes, and the fingerprint matches it
I.9	every record, event, fingerprint and model is signed and verifies against the key the seed names; the lineage chain is unbroken; with --strict, every signer is trusted here
I.10	no secret key material anywhere in the seed
I.11	stored evidence and perceptual hash are what the bytes yield
I.12	the learner was trained only on attested pages whose bytes match

I.1 to I.8 say the tree is what the catalog says. I.9 to I.12 say the catalog is what its signers said, and what the bytes say.

11. Profiles

A picture can be served four ways, and only the first two count as source.

serve as	requires
photograph	captured, human, attested, current
approved illustration	generated, ratified, current
generated, labelled	generated, unratified; shown, never source
unverified image	unknown; shown only where the profile allows

	studio	lab	internal	external
max served depth	4	2	1	0
generate	yes	yes	yes	no
ratify	no	yes	yes	no
attest	no	yes	yes	no
show unknown	yes	yes	yes	no
trusted signer needed	no	no	yes	yes

No profile serves a generated picture as a photograph. No profile serves an unratified generation or an unknown picture as source. A record whose signature does not verify is not shown at all. Internal and external serve nothing as source unless its signature verifies against a key this machine trusts, and a check that did not look at trust counts as a failure.

The studio reads MPLPB_IMAGE_PROFILE and will not generate under external. Ratify and attest name the profile they run under and refuse under studio and external.

12. Write-backs

Retire moves a current page and its bytes, unchanged, to `_log/superseded/`, off the wall, and may name a current successor.

Ratify applies to a current, unratified generated picture. It names who, stamps when, sets origin ratified at depth 0, and leaves kind generated.

Attest applies to a current unknown picture. It names who, stamps when, and requires a statement in the attester's own words of what the picture is and how they know. It sets kind captured, origin human, depth 0. It refuses any picture that matches a known generation, exactly or perceptually: the evidence wins. It refuses a picture generated only by declared metadata unless the attester asks to override (`--override-declared`). An override copies the declared signals into the signed record and the attest event, adds a conflict to the page, and is checked by I.5. A byte-identical copy of such a capture inherits it; no other picture does.

Rescan re-derives every page's stored evidence from its bytes with the current reader and re-signs what changed. A generated page whose only basis was a declaration the reader no longer finds becomes unknown, and so does any page generated only by matching it. It is how a seed written by an earlier reader is brought forward; until it is run, I.11 reports the difference.

All four are signed, logged with the same who and when as the record, and checked against each other by I.5.

13. Teaching analogue

Smart Local turns I don't know into a taught page. Image MPLPB does not. A session may not type this is a photo of X onto a file and receive a document ID that makes the claim look considered. The only route to captured is a person's name and their own sentence, signed; and the only thing a model may write onto a card is a score that says it is a score.

14. Cost

MPLPB-COST-012 spends structure at authoring time so retrieval stays ordinary. Here the authoring event is the generation or the import. `C_s` is one page, one signature, one evidence read and one hash per picture. `C_i` is a JSON catalog, a lineage log and a manifest. Validation re-reads every picture, so its cost grows with the wall; that is the price of I.11. `C_r` for a later agent is read the card instead of inferring origin from pixels.

Attestation is human attention, and Swarm MPLPB §9.2 applies unchanged: it does not amortize. A wall that needs many captures attested needs many people, and a population that attests without looking has converted the check into a formality (FM-S9).

15. What this does not claim

It does not detect generated pictures. It records evidence and refuses to invent more. It cannot tell a real photograph from a generated picture with its metadata stripped; it keeps such a picture unknown. It does not validate C2PA signatures. A signature proves which key signed a record, not that the signer told the truth. Origin depth is not a reliability score, and structural validity is not aesthetic quality or prompt fidelity. The learner's numbers here come from synthetic pictures and transfer to nothing. The pure-Python Ed25519 is correct against the RFC vectors and not constant-time. Structure has not been shown to beat a timestamped outputs/ folder.

16. Falsifiers

- F1. A stranger given only site/ cannot tell generated, captured, unknown, and retired pictures apart.
- F2. A copy that validates here fails on another machine.
- F3. ask() returns a caption for a query that hits no page.
- F4. A generated picture reaches captured, or depth 0 without a ratification record, and still validates.
- F5. Storing a picture changes its bytes.
- F6. Any profile serves a generated picture as a photograph, or an unratified or unknown picture as source.
- F7. Flat retrieval over filenames matches ask() on correct refusals.
- F8. A resized or re-encoded copy of a studio generation imports as unknown at a material rate.
- F9. A record edited without the signing key validates.
- F10. The learner's advisory score changes a picture's kind, or a non-attested page becomes a training label.
- F11. On real pictures, the learner's content-only accuracy is near chance, or its format-only accuracy explains all of its result.
- F12. Free text a person wrote into a file (a caption, comment, title or author) makes the picture declared generated, or a declared-only generation cannot be corrected by a logged, named attestation.

The reference suite is 93 standard-library unittest cases, including a fuzz pass over the evidence parser and an end-to-end run of the studio against a local fake Images API. It bears on F2 (a copied seed validates at a new path, and PNG hashes agree with and without Pillow), F3, F4 (relabelling with and without a key, forged ratification, forged attestation, forged evidence), F5, F6, F8 (half-size re-encoded copies of synthetic pictures), F9 (catalog, lineage, fingerprint, signer registry and learner tampering), F10, and F12 (captions in five fields, a JPEG comment, the override and its removal, and a rescan of a seed written by the v1 reader). It also checks the Ed25519 code against RFC 8032 §7.1 vectors 1 to 3. It bears on the structural half of F1. It does not bear on F7 or F11.

F11 needs a real set: a few hundred generations from several current models and a few hundred photographs from several cameras, attested by their owners, with the same pictures also stripped of metadata and re-encoded, and a probe author who did not choose the training pictures. F8 needs the same real set run through common platforms' resizing. Until those are run, the learner is a mechanism, not a result.

17. Commands

```
python3 -m mplpb_image init site
```

```

python3 studio/server.py
python3 -m mplpb_image validate site [--strict]
python3 -m mplpb_image check site picture.jpg
python3 -m mplpb_image import site picture.jpg --title ""
python3 -m mplpb_image attest site IMG-... --who "Name" --statement "" [--override-declared]
python3 -m mplpb_image rescan site
python3 -m mplpb_image ratify site IMG-... --who "Name" [--profile lab]
python3 -m mplpb_image retire site IMG-... --reason "" [--successor IMG-...]
python3 -m mplpb_image learn site
python3 -m mplpb_image ask site "harbour"
python3 -m mplpb_image cite site IMG-...
python3 -m mplpb_image profile check site IMG-... --name external
python3 -m mplpb_image key show | list | trust PUBLIC_KEY --name ""
python3 -m unittest discover -s tests

```

Exit codes: validate 1 on any failure; ask 4 on not in this corpus; profile check 3 when the picture would not be served as source.

18. Changes in v2

v1 searched free text for generator names, so a caption could label a photograph as declaring its own generation, and attest refused every generated picture, so that label could not be removed. v2 reads generator names only from software fields (§6), lets a named person override a declaration-only generation on explicit request (§3.1, §12), adds rescan to bring older seeds forward, and adds F12. Records written by v1 keep their signatures; the override field is signed only where it is present. A v1 seed whose pictures carry captions will fail I.11 under v2 until rescan is run, which is the check doing its job.

19. Conclusion

The toy hangs pictures. The seed remembers which were made and which were taken, and refuses to decide the second from anything a file says about itself. A copy that has been resized comes home to its card. A record that has been edited without the key stops validating. A key nobody trusts still signs validly and is still refused where it matters. The learner learns, says what it learned from, and is not allowed to write.

What it cannot do is tell a real photograph from a clean generated picture with nothing attached. It says unknown. That is the honest answer, and the design is arranged so that the honest answer is the one that survives.

Mitchell D. McPhetridge September 2026